

Question 1: Conceptual Understanding (CO1)

Task:

Explain the **Shortest Job First (SJF) CPU scheduling algorithm**.

Lab Questions:

- a) Define the **SJF Non-Preemptive scheduling algorithm**.
- b) Write the mathematical formulas for:
 - Waiting Time (WT)
 - Turnaround Time (TAT)
 - Average Waiting Time (AWT)
 - Average Turnaround Time (ATAT)
- c) Explain why SJF generally produces a **lower average waiting time** compared to FCFS.

Question 2: Manual Scheduling Calculation (CO1)

Consider the following processes:

Process Arrival Time (AT) Burst Time (BT)

P1 0 6

P2 1 8

P3 2 7

P4 3 3

Tasks

1. Draw the **Gantt Chart** for **SJF Non-Preemptive scheduling**.
2. Calculate the following for each process:
 - o Completion Time (CT)
 - o Turnaround Time (TAT)
 - o Waiting Time (WT)

3. Compute:

- o Average Waiting Time (AWT)
- o Average Turnaround Time (ATAT)

Question 3: C Programming Implementation (CO2)

Task:

Write a **C program** to implement the **Shortest Job First (SJF) Non-Preemptive CPU Scheduling algorithm**.

Program Requirements

Your program should:

1. Take **number of processes (n)** as input.
2. Take **Arrival Time (AT)** and **Burst Time (BT)** for each process.
3. Sort processes based on **Shortest Burst Time**.
4. Calculate:
 - o Completion Time (CT)
 - o Turnaround Time (TAT)
 - o Waiting Time (WT)
5. Display results in the following format:

Process	AT	BT	CT	TAT	WT
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6. Calculate and print:

- Average Waiting Time
- Average Turnaround Time

Question 4: Algorithm Analysis (CO3)

Using the following processes:

Process AT BT

P1 0 6

P2 1 8

P3 2 7

Process AT BT

P4 3 3

Tasks

1. Compute scheduling results using **FCFS algorithm**.
2. Compute scheduling results using **SJF Non-Preemptive algorithm**.
3. Compare both algorithms in terms of:
 - o Average Waiting Time
 - o Average Turnaround Time
4. Answer the following:
 - a) Which algorithm provides **better performance** for this case?
 - b) Explain the **reason for the difference** in results.